

Jim L. Zhang

EDUCATION

Stanford University, School of Medicine	Stanford, California
Ph.D. Candidate – Department of Structural Biology	2022 - Present
Rice University, Wiess School of Natural Sciences	Houston, Texas
B.S. in Biochemistry and Cell Biology – <i>summa cum laude</i>	2018-2022
Honors: Distinction in Research and Creative Work, President's Honor Roll, Phi Beta Kappa, Honors Thesis	GPA: 3.99/4.00
M.S. in Biochemistry and Cell Biology	2020-2022
Thesis: “ <i>Structure-function studies of the nematode-infecting Orsay virus</i> ”	GPA: 4.00/4.00

RESEARCH

Feng Laboratory – Stanford School of Medicine, Palo Alto, CA	
<i>Graduate Student Researcher</i>	October 2022 – Present
Feng Laboratory – Stanford School of Medicine, Palo Alto, CA	
<i>Rotation Student</i>	July 2022 – August 2022
ADVANCE Summer Institute – Stanford School of Medicine, Palo Alto, CA	
<i>Research Fellow</i>	June 2022 – August 2022
Undergraduate Honors Research in Biochemistry and Cell Biology, Houston, TX	
<i>Thesis: <u>Structural studies of a filamentous double-stranded RNA virus</u></i>	August 2021 – June 2022
<i>Advisor: Yizhi J. Tao</i>	
Advanced Biosciences Summer Research Institute, Houston, TX	
<i>Research Fellow and Student Mentor</i>	June 2020 – August 2021
National Science Foundation - REU in Multi-Scale Biomolecular Networks, Houston, TX	
<i>Research Fellow in Molecular Virology</i>	June 2019 - August 2019
B.S.-M.S.-Ph.D. Candidate in Biochemistry and Cell Biology, Houston, TX	
<i>Thesis: <u>Structure-function studies of the nematode-infecting Orsay virus</u></i>	March 2019 – June 2022
<i>Advisor: Yizhi J. Tao</i>	
Tao Laboratory – Rice University, Houston, TX	
<i>Researcher</i>	January 2019 – June 2022

TEACHING

Department of Biochemistry and Cell Biology at Rice University, Houston, TX	
<i>BIOS 302: Biochemistry II Teaching Assistant</i>	January 2022 – May 2022
Wiess School of Natural Sciences at Rice University, Houston, TX	
<i>Pilot Program for Improving STEM Retention Among Underrepresented Populations</i>	August 2021 – September 2021
Department of Biochemistry and Cell Biology at Rice University, Houston, TX	
<i>BIOS 301: Biochemistry I Teaching Assistant</i>	August 2020 – December 2021
Department of Biochemistry and Cell Biology at Rice University, Houston, TX	
<i>BIOS 450/550: Viruses and Infectious Diseases Teaching Assistant</i>	January 2021 – May 2021
Office of Academic Advising at Rice University, Houston, TX	
<i>Academic Fellow in Biochemistry and Cell Biology</i>	March 2020 – May 2022
Department of Chemistry at Rice University, Houston, TX	
<i>CHEM 211: Organic Chemistry I Teaching Assistant</i>	June 2019 – August 2019

Breakthrough Collaborative , Houston, TX <i>9th Grade Teaching Fellow in Biology; Department Chair of Science</i>	May 2018 – August 2018
Breakthrough Collaborative , Houston, TX <i>8th Grade Teaching Assistant in Chemistry and Algebra I</i>	May 2017 – August 2017
EXTRACURRICULARS AND SERVICE	
Graduate Life Office at Stanford University , Palo Alto, CA <i>Community Associate</i>	May 2023 – Present
Structural Biology and Biophysics Graduate Programs at Stanford University , Palo Alto, CA <i>Social Media Manager</i>	February 2023 – Present
Department of Structural Biology at Stanford University , Palo Alto, CA <i>Diversity, Equity, and Inclusion Committee Member</i>	September 2022 – Present
Project SHORT , USA <i>Pre-Grad Mentor</i>	July 2022 – Present
Structural Biology Graduate Program at Stanford University , Palo Alto, CA <i>Admissions Committee</i>	December 2023 – May 2024
Structural Biology and Biophysics Graduate Programs at Stanford University , Palo Alto, CA <i>Chair - Graduate Student Recruitment</i>	August 2023 – May 2024
Stanford Biosciences Student Association , Palo Alto, CA <i>NSF-GRFP Peer Mentor</i>	August 2023 – December 2023
Structural Biology and Biophysics Graduate Programs at Stanford University , Palo Alto, CA <i>Recruitment Committee Member</i>	September 2022 – May 2024
Community College Outreach Program , Palo Alto, CA <i>Science Small Group Mentor</i>	August 2022 – December 2022
Office of Academic Advising at Rice University , Houston, TX <i>Peer Academic Advisor</i>	February 2020 – May 2022
Office of First Year Programs at Rice University , Houston, TX <i>Orientation Week Coordinator – Will Rice College</i>	December 2019 – September 2020
Office of Student Success Initiatives at Rice University , Houston, TX <i>First-Generation and Low-Income Student Ambassador</i>	September 2019 – May 2020
Rice Thresher , Houston, TX <i>News Writer</i>	September 2018 – May 2019
Rice Catalyst , Houston, TX <i>Popular Science Writer and Discoveries Blogger</i>	September 2018 – May 2019
Breakthrough Collaborative , Houston, TX <i>High School Student Mentor</i>	August 2018 – May 2019

PRESENTATIONS AND CONFERENCES

Biophysical Society 2026 – Annual Meeting, San Francisco, CA

Selected Speaker; Flash Talk – “TMEM65 reconstitutes mitochondrial Na⁺/Ca²⁺ exchange”

Upcoming - February 2026

Biophysical Society 2026 – Annual Meeting, San Francisco, CA

Poster – “TMEM65 reconstitutes mitochondrial Na⁺/Ca²⁺ exchange”

Upcoming - February 2026

Department of Molecular and Cellular Physiology – 2025 Retreat, Stanford, CA

Speaker – “Molecular mechanisms of mitochondrial Ca²⁺ transport”

September 2025

Department of Structural Biology Trainee Talk, Stanford, CA

Speaker – “Structural basis of ER acetyl-CoA import”

March 2025

Rice University Shapiro Showcase, Houston, TX

Selected Speaker – “Visualizing a Virus at Molecular Resolution”

April 2022

Rice IBB Summer Research Symposium, Houston, TX

Poster - “Virus-like Particles from a Double-stranded RNA Virus with Filamentous Capsid Morphology”

August 2021

Rice Undergraduate Research Symposium, Houston, TX

Speaker - “Structural and Functional Studies of a Covalently Linked Viral Fiber”

April 2021

Cellular and Molecular Biophysics Conference, Houston, TX

Speaker - “Cryo-EM Reconstruction of a Covalently Linked Viral Fiber”

December 2020

BioSciences Summer Research Institute Symposium, Houston, TX

Speaker - “A Preliminary Cryo-EM Model of the Orsay Spike Protein CP-δ”

August 2020

Rice Undergraduate Research Symposium, Houston, TX

Poster - “The Structure and Entry Mechanism of the Nematode Virus Orsay”

April 2020

Emerging Trends in Cellular and Molecular Biophysics, Houston, TX

Poster - “Native-Condition Structural Studies of the Nematode Virus Orsay”

December 2019

Rice REU Research Symposium, Houston, TX

Poster - “A Purification Scheme Enabling Native-Condition Structural Studies of the Nematode Virus Orsay”

October 2019

Rice IBB Summer Research Symposium, Houston, TX

Poster - “An Iodixanol-based Density Gradient Isolates Orsay Virions with Substantial Yield and Purity”

July 2019

PUBLICATIONS

1. Guo, YR, Fan, Y, Zhou, Y, Jin, M, **Zhang, JL**, Jiang, H, Holt, MV, Wang, T, Young, NL, Wang, D, Zhong, W, & Tao, YJ. 2020. Orsay CP-δ adopts a novel β-bracelet structural fold and incorporates into virions as a head fiber. *Journal of Virology*.
2. **Zhang, JL***, Pan, J*, Zhou, Y, Fan, Y, Guo, YR, Zhong, W, & Tao, YJ. (*In Preparation*). The structure of a covalently linked pentameric viral fiber revealed by cryo-electron microscopy reconstruction. *Equal contribution.
3. Chen, H, Fan, Y, **Zhang, JL**, Zhou, Y, Zhong, W, & Tao, YJ. (*In Preparation*). The crystal structure of the C-terminal domain of the Orsay virus CP-δ head fiber.
4. **Zhang JL***, Chang YC*, Lai PH, Yeh H, Tsai CW, Huang YL, Foulon N, Liu TY, Lee IH, Xu Y, Rao B, Tu YC, Feng L, and Tsai MF. 2025. TMEM65 functions as the mitochondrial Na⁺/Ca²⁺ exchanger. *Nature Cell Biology*. *Equal contribution. *Featured in Spotlight article by Cell Metabolism (PMID: 41061666)*

FELLOWSHIPS AND GRANTS

1. Barry M. Goldwater Foundation. Goldwater Scholar. March 2021 – May 2022, \$7,500.
2. Rice University. George J. Schroepfer Jr. Summer Undergraduate Research Fellowship. May 2021 – August 2021, \$5,000.
3. National Science Foundation. Graduate Research Fellowship Program. September 2022 – September 2027, \$138,000.

4. National Institutes of Health. Ruth L. Kirschstein National Research Service Award – Predoctoral. “Molecular basis of mitochondrial calcium extrusion - a key target in ischemic heart disease” (1F31HL182219-01 [5th Percentile]), September 2025 – September 2027, \$99,076.
5. Blavatnik Family Foundation. Blavatnik Fellowship. December 2025 – December 2026.

HONORS AND AWARDS

Shapiro Prize , Houston, TX	April 2022
George J. Schroepfer Jr. Award for Excellence in Undergraduate Research in Biochemistry , Houston, TX	April 2022
Worden Endowed Award , Houston, TX	April 2022
Rice Excellence in Academic Advising Award , Houston, TX	April 2022
Rice IBB Research Symposium , Houston, TX <i>Outstanding Poster Presentation</i>	August 2021
Rice Undergraduate Research Symposium , Houston, TX <i>Outstanding Oral Presentation</i>	April 2021
James Street Fulton and Edythe King Fulton Endowed Prize , Houston, TX	December 2020
Willis Service Award , Houston, TX	November 2020
BIOC 211 Student Research Presentations , Houston, TX <i>Outstanding Poster Presentation</i>	November 2019
Rice REU Research Symposium , Houston, TX <i>Outstanding Poster Presentation</i>	November 2019
Comcast Leaders and Achievers Scholar , Houston, TX	June 2018
HISD Board of Education Award , Houston, TX <i>Outstanding Student</i>	May 2018